U.S. Memory-Chip Imports: Sourcing, Product Mix, and Tariffs

1. Key Findings

1. **Import value from China increased, while China's share of U.S. imports declined.** From January through September 2026, import value from China increased 95.3% year over year, while its share fell from 3.6% to 1.7%, a decline of 1.9 percentage points. In the Shapley decomposition, the change in China's own import value contributed +2.0 percentage points, while changes in all other origins contributed -3.9 percentage points. **[Derived]** — Source: `tables/share_decomposition.csv`
2. **Import-value growth did not move with reported quantity.** From 2024 to 2025, World import value increased 22.5%, reported quantity decreased 25.0%, and the unit-value proxy increased 63.3%, from $1.25 per unit to $2.04 per unit. A change in import value therefore cannot be treated as a change in physical quantity. **[Derived]** — Source: `tables/origin_change.csv`
3. **Origin concentration increased as China's share declined.** From 2024 to 2025, China's share fell from 4.4% to 3.4%, while HHI increased from 2,066 to 2,179 and the top-three-origin share increased from 70.6% to 74.0%. These results are more consistent with reallocation toward leading non-China origins than with broad diversification. **[Derived / Inference]** — Source: `tables/scope_series.csv`
4. **A few HTS10 categories dominate the product mix, and China exposure differs across categories.** In the 2025 World composition, `DRAM - >1 Gb` accounted for 35.8%, `EEPROM` for 33.7%, and `Other` for 20.3%. China supplied 6.1% of U.S. imports within `EEPROM`, compared with 1.4% within `DRAM - >1 Gb`. **[Measured / Derived / Inference]** — Source: `tables/product_hierarchy_matrix.csv`
5. **The data show a sharp break around the 2018 policy date, but establish only timing alignment.** From June to August 2018, China's share increased 12.2 percentage points. From August to September 2018, its share fell 23.3 percentage points, while the realized duty rate rose from 0% to 19.7%. The Section 301 List 2 effective date falls within this window, but these data do not establish causation. **[Derived]** — Source: `tables/share_decomposition.csv`
6. **Statutory treatment and realized duty must be read separately.** As of 2026-11-09, ordinary consumption entries from China in all nine HTS10 categories had an MFN rate of Free plus a 50% additional duty under `9903.91.05`. From January through September 2026, the realized duty rate was 40.6% for China and 0.7% for World. **[Measured / Derived]** — Source: `tables/tariff_treatment_snapshot.csv`, `tables/origin_change.csv`

2. Tariff Timeline and Treatment Snapshot

Effective date	Layer	Rate change	HTSUS / Executive Order / Federal Register	Treatment in this study
2017-01-01	MFN base rate	Free	HTS 8542.32.00	Base layer
2018-08-23	Section 301 List 2	+25%	`9903.88.02`; 83 FR 40823	Applies
2025-01-01	Section 301 semiconductor modification	25% to 50%	`9903.91.05`; 89 FR 76581	Replaces the earlier Section 301 layer; not stacked twice
2025-02-04	Synthetic-opioid policy IEEPA layer	+10%	`9903.01.20`; EO 14195; 90 FR 9121; CBP 90 FR 9038	Applies
2025-03-04	Synthetic-opioid policy IEEPA layer	10% to 20%	`9903.01.24`; EO 14228; 90 FR 11463; CBP 90 FR 11426	Applies
2025-11-10	Synthetic-opioid policy IEEPA layer	20% to 10%	`9903.01.24`; EO 14357; 90 FR 50725	Applies
2026-02-20	IEEPA layer ended	10% to 0%	EO 14389; 91 FR 9437	Ended; Section 301 remains

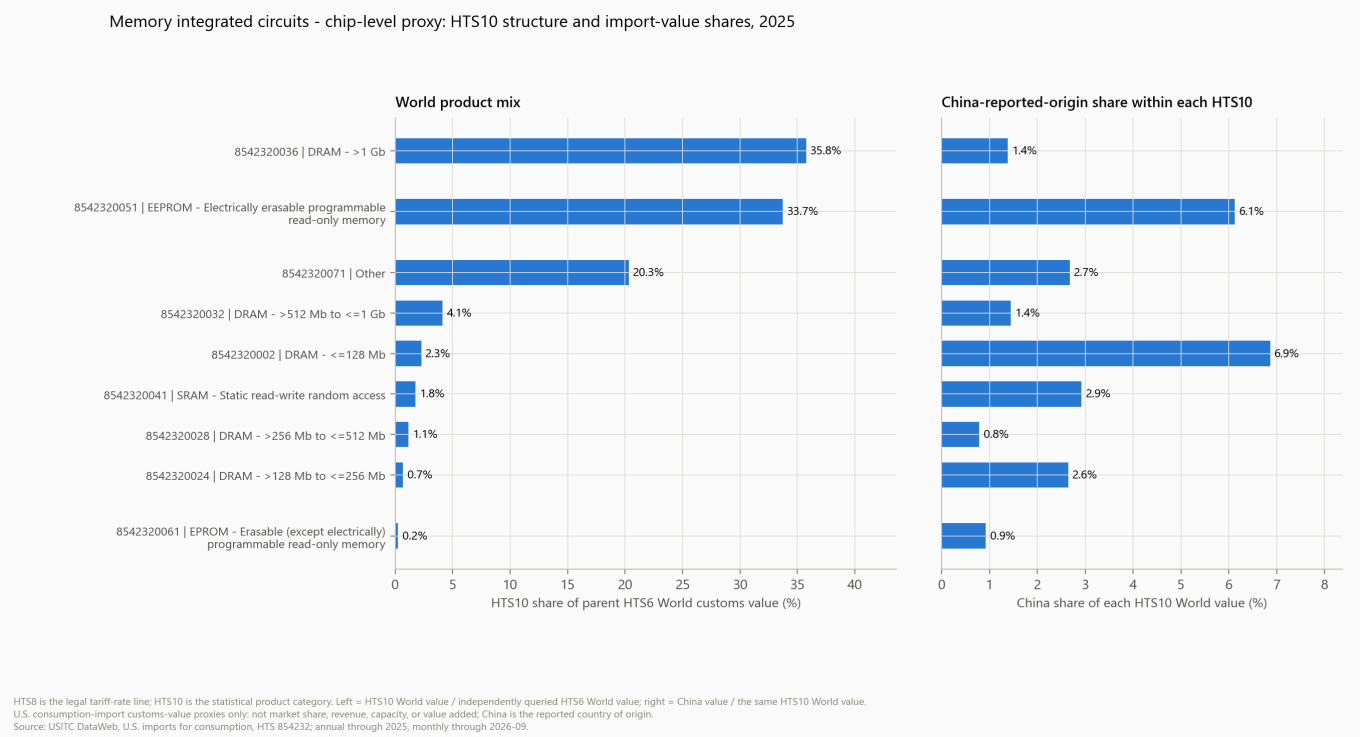
[Measured] — Source: `tables/tariff_timeline.csv`

The synthetic-opioid supply-chain policy is the legal rationale for this broad origin-based IEEPA layer. It does not describe memory integrated circuits as opioid-related goods. The layer covered most goods originating in China and Hong Kong. In the current snapshot, the IEEPA layer has ended, and ordinary entries in all nine HTS10 categories have an MFN rate of Free plus a 50% additional duty under `9903.91.05`. **[Measured]** — Source: `tables/tariff_timeline.csv`, `tables/tariff_treatment_snapshot.csv`

Primary legal sources: 2018 Section 301 List 2, 2025 Section 301 semiconductor modification, EO 14195, EO 14228, EO 14357, and EO 14389.

3. Body: Scale, Origin Mix, and HTS10 Product Structure

The study hierarchy is `HTS6 854232 > HTS8 85423200 > nine HTS10 statistical categories`. The scope contains only one HTS8 legal-rate line. The body therefore shows the HTS10 structure directly instead of adding a separate hierarchy diagram. **[Measured]** — Source: `tables/product_hierarchy_matrix.csv`



[Measured / Derived] — Chart data: `charts/12_product_mix.csv`; full matrix: `tables/product_hierarchy_matrix.csv`.

HTS10	Readable category (English)	Official HTS description (English)	World import value (USD million)	Share of HTS6 (%)	China share within HTS10 World (%)
`8542.32.00.36`	DRAM - >1 Gb	Over 1 gigabit	800.9	35.8	1.4
`8542.32.00.51`	EEPROM - Electrically erasable programmable read-only memory	Electrically erasable programmable read-only memory (EEPROM)	755.6	33.7	6.1
`8542.32.00.71`	Other	Other	455.3	20.3	2.7
`8542.32.00.32`	DRAM - >512 Mb to <=1 Gb	Over 512 megabits but not over 1 gigabit	92.1	4.1	1.4
`8542.32.00.02`	DRAM - <=128 Mb	Not over 128 megabits	50.8	2.3	6.9
`8542.32.00.41`	SRAM - Static read-write random access	Static read-write random access (SRAM)	39.6	1.8	2.9

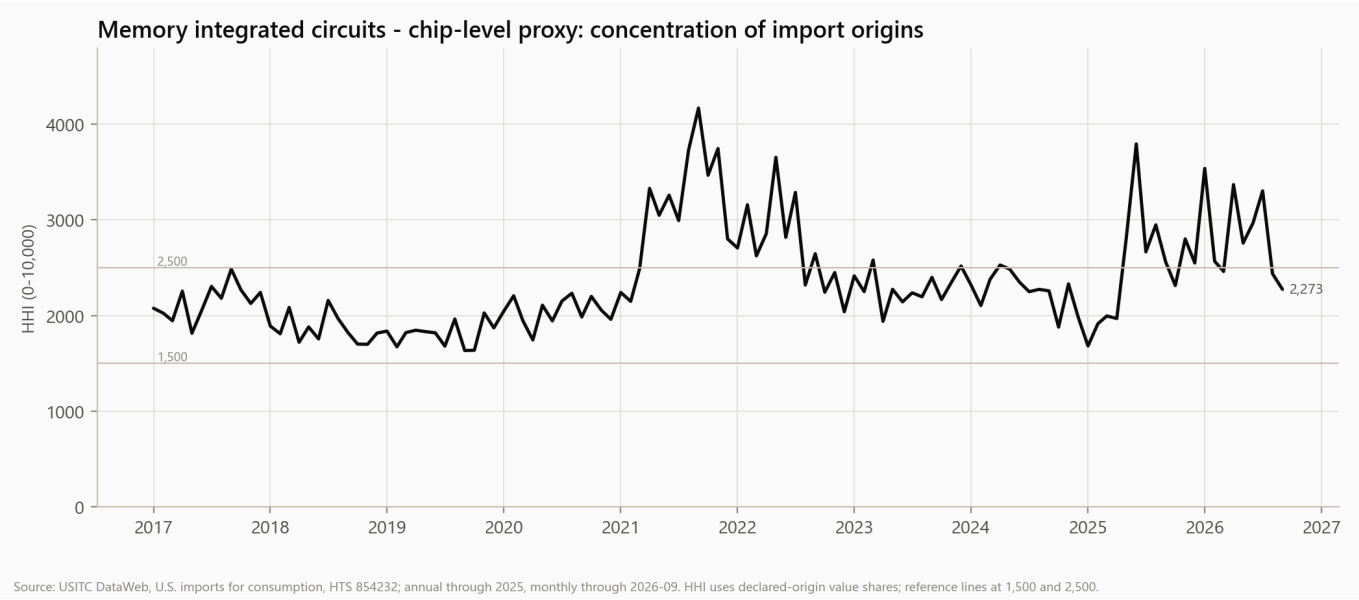
HTS10	Readable category (English)	Official HTS description (English)	World import value (USD million)	Share of HTS6 (%)	China share within HTS10 World (%)
`8542.32.00.28`	DRAM - >256 Mb to <=512 Mb	Over 256 megabits but not over 512 megabits	25.4	1.1	0.8
`8542.32.00.24`	DRAM - >128 Mb to <=256 Mb	Over 128 megabits but not over 256 megabits	15.3	0.7	2.6
`8542.32.00.61`	EPROM - Erasable (except electrically) programmable read-only memory	Erasable (except electrically) programmable read-only memory (EPROM)	5.3	0.2	0.9

[Measured / Derived] — Source: `tables/product_hierarchy_matrix.csv`

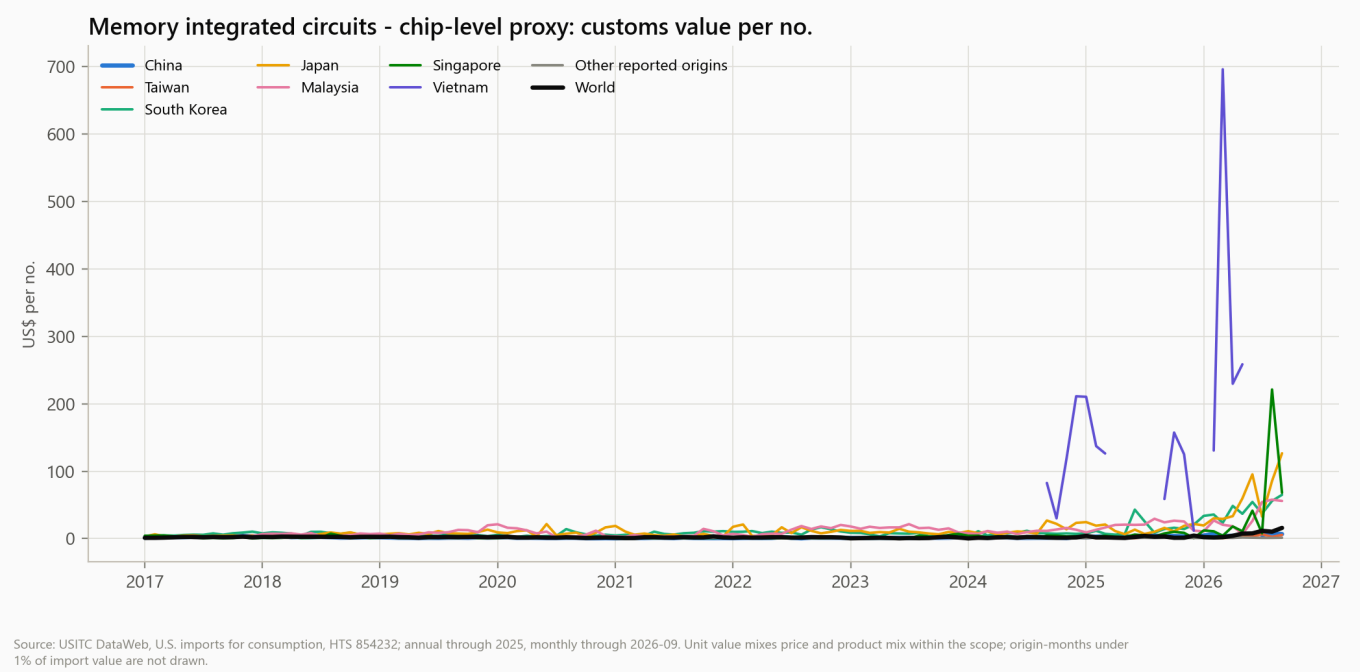
Mb and Gb in the table are bit-capacity units, not MB or GB. `Readable category (English)` is a compact English label generated from the official hierarchy, while the adjacent column preserves the official English description from the 2025 HTS Basic Edition. The five DRAM capacity categories are identified through the official blank-tariff-line superior heading `Dynamic read-write random access (DRAM)`, not inferred from the capacity wording alone. The matrix measures the customs-value composition of U.S. imports for consumption. It does not measure global production capacity or U.S. market share. Official 2025 HTS Basic JSON

From January through September 2026, South Korea, Japan, and Singapore gained 8.6, 2.8, and 8.8 percentage points of import share, respectively. Import value from Taiwan, Malaysia, and Vietnam still increased, but their shares fell 3.5, 9.7, and 2.1 percentage points, respectively. China's share fell 1.9 percentage points. Import value and import share can therefore move in opposite directions. **[Derived]** — Source: `tables/origin_change.csv`

Over the same period, World import value increased 310.1%, reported quantity increased 47.1%, and the unit-value proxy increased 178.8%. This divergence requires separate readings of quantity, unit value, and HTS10 composition. The current HTS10 shares are a 2025 cross-section and cannot measure year-to-year product-mix changes. **[Derived]** — Source: `tables/origin_change.csv`, `tables/product_hierarchy_matrix.csv`



[Derived] — Chart data: `charts/07_concentration.csv`.



[Derived] — Chart data: `charts/08_unit_value.csv`.

4. Questions Worth Asking

1. What stage of the memory-industry cycle, capacity transition, and end-market demand applied during the study window? Answering this question requires the user to authorize optional external industry research.
2. How does reported China-origin exposure relate to specific companies, wafer fabrication plants, assembly and test facilities, and supply agreements?
3. How did the HTS10 product composition change from 2024 to 2025 and from 2025 to 2026? Can the change in the unit-value proxy be separated into within-category movement and shifts between HTS10 categories? **[Unknown]** — Source: `tables/product_hierarchy_matrix.csv`
4. Which entry mix, exclusions, declaration dates, or special treatments explain the difference between the 50% statutory treatment and China's 40.6% realized duty rate? **[Unknown]** — Source: `tables/tariff_treatment_snapshot.csv`, `tables/origin_change.csv`

5. Drill-down and Next Steps

1. **Extend the HTS10 product-mix history.** Acquire World and China data for the comparison years using the same archived HTS structure, then separate within-category unit-value movement from shifts between HTS10 categories. **[Measured]** — Source: `tables/next_steps.csv`
2. **Verify historical HTS continuity.** Complete the official HTS archive record and bridge any code splits or mergers across the study period. Until then, continue to label continuity as unverified. **[Measured]** — Source: `tables/next_steps.csv`, `analysis_manifest.json`
3. **Use a separate evidence layer for company dependence.** If the research question shifts from reported country of origin to company dependence, add company shipment, supply-chain, or corporate-disclosure data. DataWeb cannot identify corporate ownership or the origin of value added. **[Inference]** — Source: `tables/proxy_notes.csv`

6. Notes on Measures

- Source: USITC DataWeb, U.S. imports for consumption. Annual data run through 2025, and monthly data run through 2026-09. **[Measured]** — Source: `analysis_manifest.json`
- The unit-value proxy is `CONS_CUSTOMS_VALUE / CONS_FIR_UNIT_QUANT`, reported as USD per No. It is not a transaction price, company ASP, or capacity-adjusted price index. It changes with technology, capacity, packaging, specification, and product mix. **[Derived]** — Source: `tables/proxy_notes.csv`
- The realized duty rate is `CONS_CALC_DUTY / CONS_CUSTOMS_VALUE`. It is not the statutory tariff rate and does not establish that every entry used the same Chapter 99 provision. **[Derived]** — Source: `tables/proxy_notes.csv`
- China, South Korea, Taiwan, Japan, Malaysia, Vietnam, and Singapore are origins reported to U.S. Customs. They do not identify corporate ownership, the origin of value added, wafer-fabrication location, or exposure adjusted for transshipment. **[Measured]** — Source: `tables/proxy_notes.csv`
- Each HTS10 product share equals same-period HTS10 World customs value divided by the independently queried parent HTS6 World customs value. It is not U.S. market share, global production capacity, company revenue share, or shipment-volume share. **[Derived]** — Source: `tables/proxy_notes.csv`

Appendix

The body is brief.md as written (SHA-256 7a208f89f2e4...); the appendix is generated by final_report from validated files.

A. Study identity

Item	Value
Product	Memory integrated circuits - chip-level proxy (HTS6: 854232)
Scope	memory-integrated-circuits v1 scope_sha256 `e007c6aadcdb7e2d6811f1e811b721fcc0f3a18c15dc0df3fe257fd7d49680ac`
Study	memory-integrated-circuits-2017-2026 study_sha256 `f0c39241c215b60500b20259977d3c18291df059eaa3e77cfdaef99c79c2ca8a`
Plan	plan_sha256 `ad0ab40fc6154263e160e3b2765a50689a6c2ddad95d4c34ebda52aef899aed2`
Data	USITC DataWeb, U.S. imports for consumption; data through annual 2025, monthly 2026-09 (queried through annual 2025, monthly 2026-12; DataWeb has no data yet after the data end)
Approval	approved (confirmed by the user in chat and recorded by the agent's scope-approve run; their words are in the scope manifest)
HTS continuity	**not verified**
Quantity	first unit no.; second unit — (basis = first)
Analysis generated (UTC)	2026-11-09T17:49:41.968914+00:00

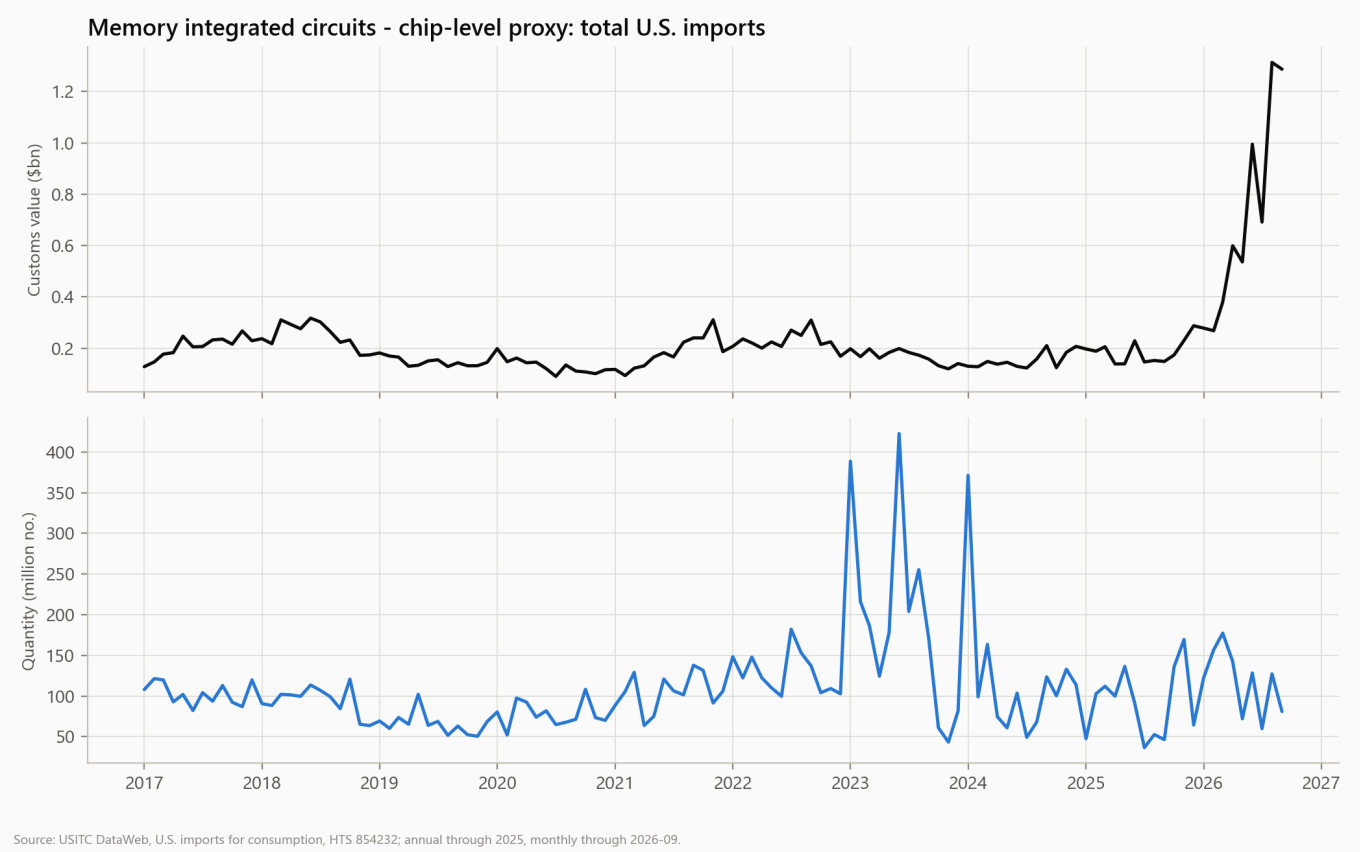
B. Data checks

Item	Value
Panel validation	passed: 7545 rows, 126 product periods, 0 errors, 0 warnings; all 10 charts have their data CSV
Origins against World	126 rows, status ok 126; largest gap: value 0.000%, quantity 0.000%, calculated duty 0.000%
Code coverage	2 code-frequency rows, 0 need review, 0 have gaps
Body brief-check	passed: 163 numeric tokens, 81 with a recognized unit, 81 found in the validated files they cite; 0 ambiguous units, 82 unitless skips; 12 files cited; 0 warnings

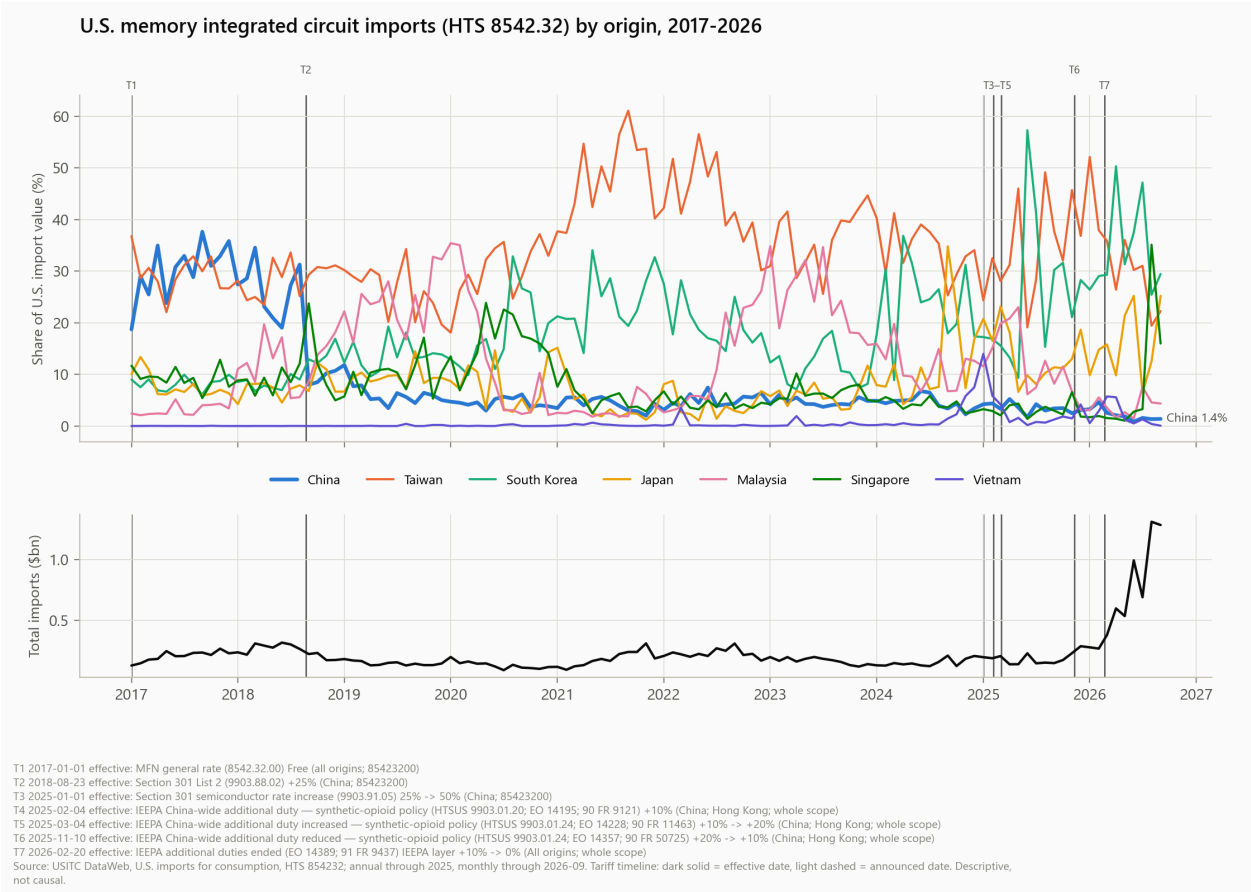
C. What this cannot show

- Country or area of origin is the origin reported to U.S. Customs; it is not value-added origin, fab location or firm dependence.
- A gaining source does not prove direct relocation from the focal origin.
- Policy-aligned timing is descriptive unless a causal design is added.
- Realized duty (calculated duty / customs value) is not the statutory tariff rate.
- The HTS tariff-treatment snapshot is non-binding research, not a CBP classification ruling or entry liquidation.
- Unit value (customs value / reported quantity) is a product-mix proxy, not a transaction price, company ASP or per-capacity price.
- HTS10 product share is a share of U.S. import customs value inside the parent HTS6, not market share or global capacity.

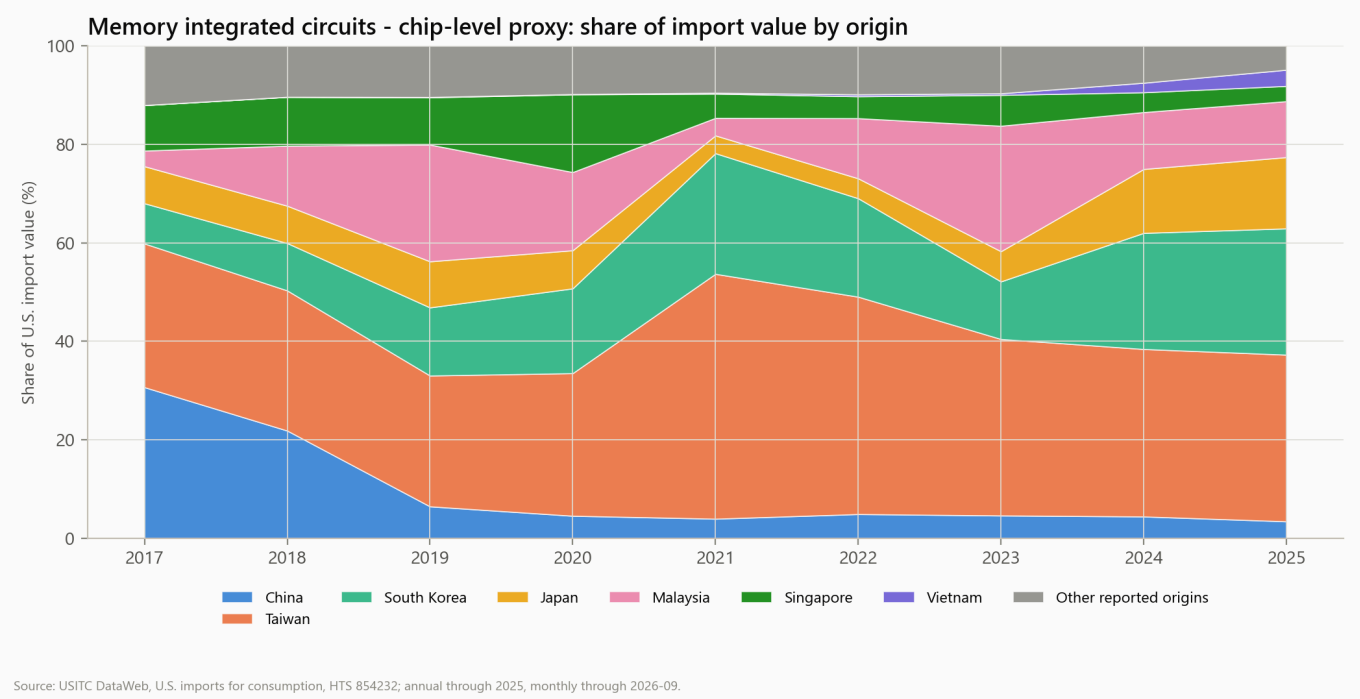
D. Other charts



01_scope_trend · Chart data: `charts/01_scope_trend.csv`

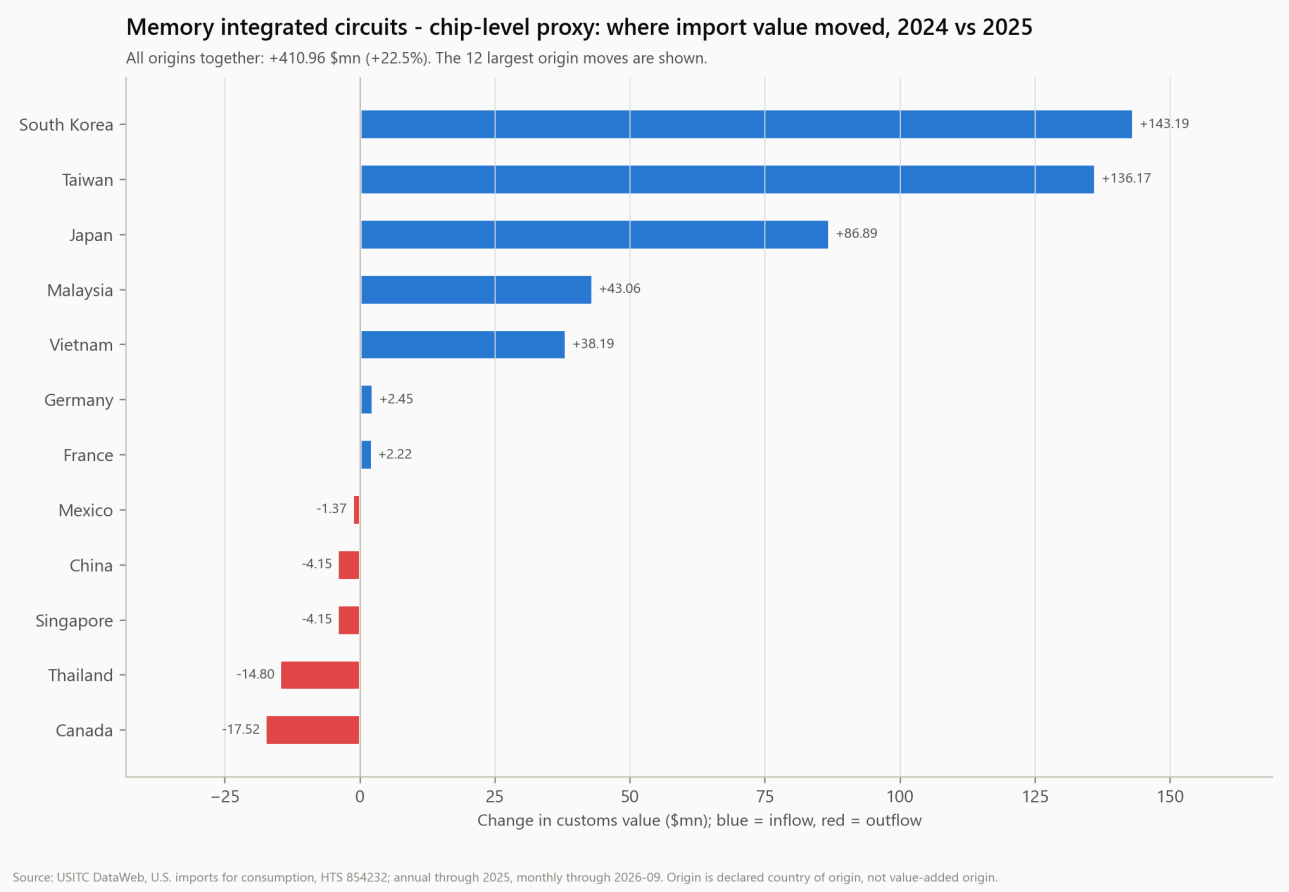


03_event_path · Chart data: `charts/03_event_path.csv`



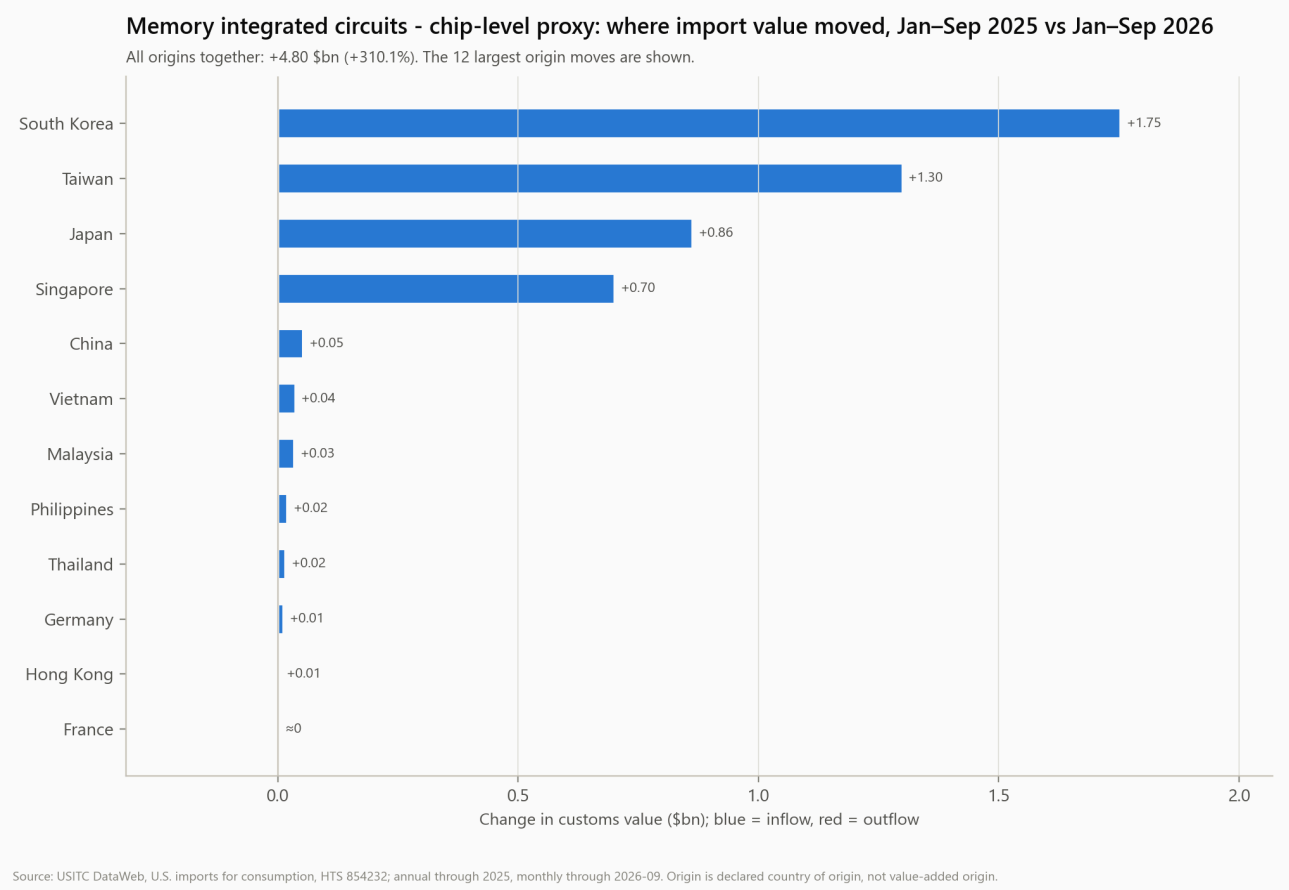
Source: USITC DataWeb, U.S. imports for consumption, HTS 854232; annual through 2025, monthly through 2026-09.

04_origin_shares · Chart data: `charts/04_origin_shares.csv`

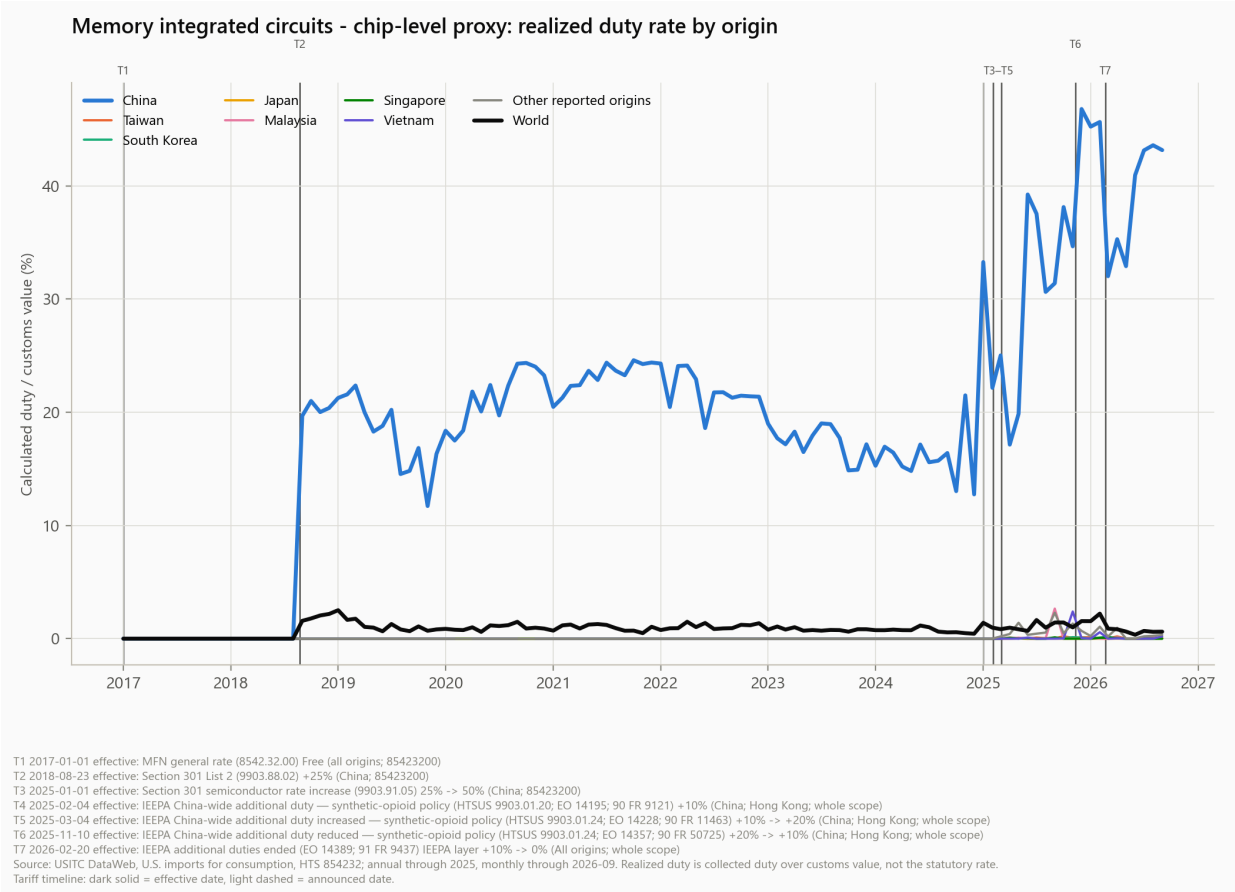


Source: USITC DataWeb, U.S. imports for consumption, HTS 854232; annual through 2025, monthly through 2026-09. Origin is declared country of origin, not value-added origin.

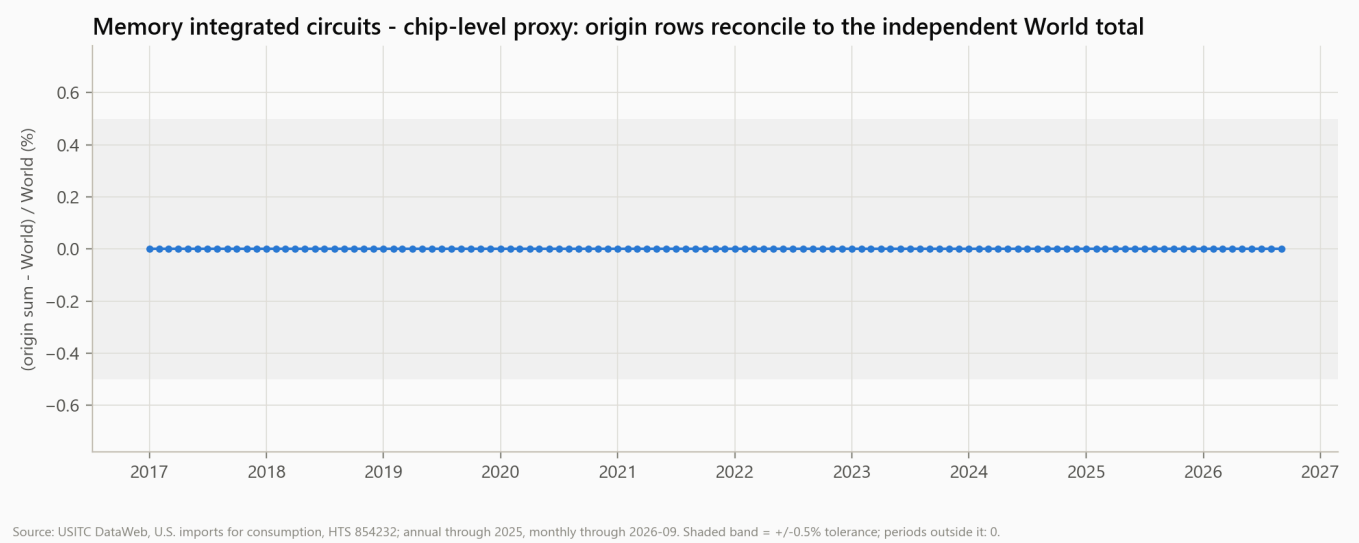
05_origin_change · Chart data: `charts/05_origin_change.csv`



06_ytd_comparison · Chart data: `charts/06_ytd_comparison.csv`



09_realized_duty · Chart data: `charts/09_realized_duty.csv`



10_world_check · Chart data: `charts/10_world_check.csv`

E. Files and hashes

File	Path	SHA-256 (first 12)
brief	`brief.md`	`7a208f89f2e4`
anomaly_flags	`tables/anomaly_flags.csv`	`8da1d776838a`
coverage_audit	`tables/coverage_audit.csv`	`c37e8767053a`
display_series	`tables/display_series.csv`	`2e621e2d2b6e`
drilldown_menu	`tables/drilldown_menu.csv`	`a6ec44dc3d5c`
event_metrics	`tables/event_metrics.json`	`12a5a9942315`
findings	`findings.json`	`be565d3028c2`
focal_split	`tables/focal_split.csv`	`27f75babddd1`
mention_list	`tables/mention_list.csv`	`e6f7d1719afc`
monthly_panel	`tables/monthly_panel.csv`	`89d6ed06352e`
next_steps	`tables/next_steps.csv`	`190ffeba2c30`
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window_candidates	`tables/window_candidates.csv`	`27bb64f100b0`
world_check	`tables/world_check.csv`	`5fad5c02caa8`